FastDedup

A fast and memory efficient tool for reads deduplication

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Abstract — PCR duplicate removal is a critical first step in high-throughput sequencing pipelines, yet existing tools struggle with speed, memory, or correctness at modern dataset scales. We present `FastDedup`, a Rust-based FASTX deduplicator that transform each read or read pair to a compact `xxh3` hash fingerprint, drastically reducing memory usage, and binding most of the execution time to disk I/O. Benchmarked against six competing tools on synthetic human WGS datasets up to 300 million reads, `FastDedup` consistently leads on paired-end data, running more than 10 times faster than `fastp` while using a fraction of `fastp` RAM usage overall. It also outperforms all tools on uncompressed single-end data, deduplicating a million reads in a second. We additionally report a correctness failure in `PRINSEQ++` and `clumpify`. `FastDedup` is available under the MIT License via GitHub, Bioconda, and Cargo.

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1 – Context

With the rise of New Generation Sequencing (NGS) technologies, the volume of sequencing data has increased exponentially. With this surge in data, the development of new tools for processing and analyzing sequencing reads was needed. One of the first and critical step in the analysis pipeline is the deduplication of reads, which involves identifying and removing duplicate sequences that can arise from PCR amplification or sequencing errors. This is crucial when working with DNA-seq data, but can introduce bias in RNA-seq data since the duplication could be biological in nature (e.g., highly expressed genes).

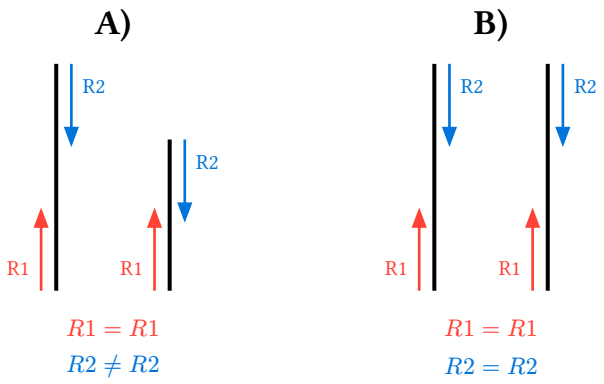


Figure 1: Identifying PCR Duplicates in Paired-End Sequencing.

Solid Black Line: Represents a single sequenced DNA fragment (height reflects length).

Red Arrow (R1): Forward read (Read 1).

Blue Arrow (R2): Reverse read (Read 2).

A) Not a duplication: Shows two distinct fragments where the start position of Read 1 is shared, but different fragment lengths lead to unique mapping positions for Read 2.

B) PCR Duplication: Shows two exact fragments where the start position of Read 1 and Read 2 is shared.

We define a fragment F, the 5′->3′ sequence as $R1$ with a length of r_1 and the 3′->5′ sequence as $R2$ with a length of r_2. We can say that $R1$ is the prefix of F, $R2$ reversed is the suffix of F and $r_1 = r_2$.

Therefore, a PCR duplicate can be defined as if F_1 and F_2 have the same starting and ending position in the genome, with the same length. In that case, $R1_1$ and $R1_2$ are the same, and $R2_1$ and $R2_2$ are the same.

This step is crucial because removing these artifacts preventing false-positive during variant calling since PCR polymerases can introduce errors during the cycles of amplification. If an error occurs early, it is propagated into multiple daughter molecules. When sequenced, these identical, error-carrying molecules will appear as multiple independent reads supporting a mutation that does not actually exist in the biological sample [1].

Deduplication also allows correcting the allele frequency bias since not all DNA molecules amplify with equal efficiency due to factors like GC content or fragment length. This uneven amplification can artificially inflate the coverage of certain alleles. And it is also improving *de novo* assembly, especially in whole-genome sequencing. Paired-end reads are mapped to estimate the order and intervening distance between contigs, duplicates can create false-positive connections between contigs or introduce conflicting connections. [2].

Before presenting our tool, we will first review the existing tools for reads deduplication and their limitations.

2 – Existing tools and limitations

2.1 : Fastp

`fastp` is a widely adopted, all-in-one FASTQ preprocessor written in C++ that performs quality control, adapter trimming, and deduplication within a single pass [3,4]. Unlike alignment-based tools that require computationally expensive mapping and coordinate sorting, `fastp` operates directly on raw sequences.

To achieve high performance, `fastp` implements a probabilistic deduplication mechanism based on Bloom filters and multiple hash arrays. For each read, or read pair, the sequence is mapped to multiple integer hash values. If all corresponding positions in the hash arrays are already marked as positive, the read is flagged as a duplicate. This streaming, single-pass architecture avoids reading and writing intermediate files to disk, drastically reducing I/O bottlenecks and resulting in ultra-fast execution speeds.

A significant advantage of `fastp` is its highly configurable memory management. Users can adjust the `--dup_calc_accuracy` parameter to scale the hash buffer size from 1 GB up to 24 GB, allowing the tool to run on both standard personal computers and High-performance Computing (HPC) clusters.

However, this probabilistic approach involves a minor trade-off in absolute precision. `fastp` has a theoretical false-positive rate. Furthermore, `fastp`'s deduplication algorithm relies heavily on the exact matching of sequence lengths and coordination regions. Consequently, applying certain quality trimming operations (such as sliding-window trimming via `--cut_front` or `--cut_tail`) prior to deduplication can interfere with the duplicate identification process, as trimmed reads may no longer perfectly match their duplicate counterparts.

2.2 : FastUniq

`FastUniq` [5] is a deterministic deduplication tool explicitly designed for paired-end (PE) short reads. By directly comparing the nucleotide sequences of read pairs, it provides an exact identification of PCR duplicates.

Algorithmically, `FastUniq` operates by loading the entirety of the paired reads into memory, sorting them based on their sequences using a merge sort algorithm andidentifiying duplicates by sequentially comparing adjacent read pairs in the sorted list.
A notable feature of its comparison logic is its ability to handle reads of varying lengths: a shorter sequence is considered identical to a longer one if it perfectly matches its 5′ prefix. If duplicates are found, the pair containing the longest reads is retained.

While `FastUniq` guarantees absolute precision for exact duplicate removal, its architectural design introduces critical resource bottlenecks. Because it relies on importing the entire dataset into RAM simultaneously using a hierarchical storage structure, its memory footprint scales linearly with the input file size. Benchmarking studies have highlighted this as a severe limitation for modern high-throughput data. For instance, processing a moderately sized dataset of approximately 69 million read pairs required over 50 GB of RAM [6].

While highly accurate, `FastUniq` is often impractical for massive datasets without access to HPC clusters. Furthermore, its scope is strictly limited to paired-end data, it does not tolerate sequence mismatches.

2.3 : Clumpify

`Clumpify`, a tool of the Java-based BBTools suite [7], introduces a algorithmic paradigm to duplicate removal based on k-mer grouping. Rather than relying on exact sequence hashing or global coordinate sorting, `Clumpify` identifies reads that share common k-mers and groups them into localized clusters or "clumps".

This spatial reorganization provides a unique secondary benefit. By physically placing highly similar sequences adjacent to one another in the output file, `Clumpify` dramatically improves standard `gzip` compression efficiency, as the compression algorithm can utilize much closer reference pointers.

A major functional advantage of `Clumpify` is its high sensitivity to optical duplicates. By parsing the flowcell tile and pixel coordinates embedded within Illumina FASTQ headers, `Clumpify` (via parameters such as `dupedist`) can explicitly distinguish between amplification-derived PCR duplicates and sensor-derived optical duplicates. For paired-end data, `Clumpify` predominantly uses the first read (R1) to determine the clump, allowing the paired read (R2) to passively follow its counterpart, which maintains file synchronization while minimizing computational overhead.

To manage memory, `Clumpify` implements a multi-phase strategy (KmerSplit and KmerSort). If the dataset exceeds available RAM, the tool can divide the data into temporary groups on the disk, sorting them individually before merging.

However, its reliance on the Java Virtual Machine (JVM) can lead to instability and high memory overhead on massive datasets. By default, high-memory tools within the BBTools suite will attempt to allocate all available system RAM [7], and JVM garbage collection overhead can cause severe bottlenecks unless strict multi-threading and heap-size limits are manually tuned.

2.4 : SeqKit

`SeqKit` [8,9] is a comprehensive, cross-platform, and ultra-fast toolkit for FASTA and FASTQ file manipulation. Written in the Go, it benefits from efficient concurrency management and the ability to produce static binaries with no external dependencies, making it highly accessible and stable across different operating systems.

For PCR deduplication, SeqKit provides the `rmdup` subcommand. This tool relies on an exact sequence matching approach using a hashing strategy. To process massive datasets without exhausting system memory, `seqkit rmdup` offers an MD5 mode (invoked via the `-m` or `--md5` flag). In this mode, the algorithm computes and stores only the MD5 digest of each sequence in the RAM rather than the full nucleotide sequence. This extreme memory efficiency allows SeqKit to maintain a very low resource footprint, even outperforming newer tools (e.g `prinseq++`) when handling vast amounts of single-end data.

Despite its speed and memory efficiency, `SeqKit rmdup` has a notable architectural limitation: it does not natively support the synchronized deduplication of paired-end reads. The `rmdup` command evaluates and processes individual files independently. It will check duplicates inside each file separately, so if $R1_1 = R1_2$ but $R2_1 \neq R2_2$, $R1_2$ in will be flagged as a duplicate and removed, while $R2_2$ will be retained, leading to loss of data and desynchronization of the paired files. While workarounds exist using other SeqKit subcommands (such as `seqkit pair` to resynchronize files), `SeqKit rmdup` is generally considered the optimal choice strictly for single-end libraries. This is why we will not use it for paired-end benchmarking in this study.

2.5 : CD-HIT-DUP

`CD-HIT-DUP`, a specialized tool within the broader CD-HIT package [10,11], is designed to identify and remove both identical and nearly-identical duplicates. Unlike tools restricted to exact string matching, `CD-HIT-DUP` is particulary robust against sequencing errors. It incorporates a greedy incremental clustering algorithm that tolerates a defined threshold of mismatches, including both substitutions and insertions/deletions.

`CD-HIT-DUP` relies on a prefix-suffix comparison approach. The tool first bins reads that share an identical prefix of k nucleotides. To accelerate this binning and sorting process, it uses a dual numeric encoding strategy, allowing prefixes of up to 27 nucleotides to be compactly represented as 64-bit integers. Within each bin, sequences are sorted by decreasing length, with the longest sequence acting as the cluster's seed. The algorithm then evaluates the suffixes of the remaining sequences against this seed, utilizing a k-mer word-based index table to statistically predict sequence identity and bypass computationally expensive full pairwise alignments when possible. For paired-end libraries, `CD-HIT-DUP` conceptually joins the read pairs into a single sequence and checks prefixes at both ends simultaneously to ensure paired-end integrity.

2.6 : PRINSEQ++

`PRINSEQ++` [12] is an optimized C++ implementation of the Perl-based `prinseq-lite` program. It is designed to perform a comprehensive suite of quality control, filtering, trimming, and reformatting tasks for genomic and metagenomic sequence data.

For sequence deduplication, `PRINSEQ++` departs from exact string comparison or sorting heuristics by implementing a probabilistic data structure known as a Bloom filter. During execution, each sequence is transformed using multiple fast, non-cryptographic hash functions, and the corresponding indices in a shared bit-array are flagged. If a newly evaluated sequence maps entirely to bits that are already set to positive, it is classified as a duplicate. This probabilistic model is highly advantageous for multi-threaded environments because it enables asynchronous read and write access to the filter with minimal thread blocking.
Additionally, `PRINSEQ++` natively supports the direct reading and writing of `gzip`-compressed FASTQ and FASTA files without the need for intermediate disk I/O, which significantly reduces hard drive storage requirements.

It also natively supports paired-end reads by processing pairs simultaneously and outputting synchronized R1 and R2 files. However, the deduplication function (`-derep`) is strictly limited to exact duplicates and does not tolerate sequencing errors. Furthermore, the probabilistic nature of the Bloom filter introduces theoretically non-zero rate of false positives. Finally, when `PRINSEQ++` is executed with multiple threads, the output sequences are not guaranteed to remain in the same order as the input files, which may require consideration if downstream tools expect strict global index synchronization.

2.7 : SAM/BAM ecosystem tools

In addition to tools that operate directly on raw FASTQ sequences, the bioinformatics ecosystem relies heavily on utilities designed for SAM and BAM formats. PCR deduplication tools like such as `samtools` (via `markdup`) [13] and Picard's `MarkDuplicates` [14].
While these tools are typically used downstream after mapping reads to a reference genome, it is technically possible to convert raw, unaligned FASTQ files directly into unmapped SAM/BAM files to process them through these pipelines. However, even without the computationally heavy step of genomic alignment, the sheer format conversion from FASTQ to SAM/BAM introduces significant and unnecessary I/O and processing overhead. Because our primary objective is to evaluate and develop a highly optimized, memory-efficient tool strictly for direct FASTQ-to-FASTQ deduplication, these format-dependent tools fall outside the scope of our comparative analysis.

3 – FastDedup: High-Performance FASTX Deduplication

The goal behind the development of `FastDedup` (or `FDedup` ; Figure 2 Figure 3) was to create a FASTX PCR deduplication tool that prioritizes maximum speed and memory efficiency. To achieve this, the tool relies on `xxh3` [15,16], a non-cryptographic hash function, to compute a unique fingerprint for each read. These fingerprints are cached in memory using `fxhash` [17], which provides a low memory footprint.

FDedup provides users with precise control over the hash collision rate while maintaining high performance. It automatically scales its hashing strategy, seamlessly switching between 64-bit and 128-bit hashes based on the estimated input sequence count and a user-defined collision probability threshold.

Specifically designed for seamless integration into OMICS pipelines, the tool features robust incremental deduplication and auto-recovery mechanisms. In the event of an interruption FDedup can safely preload existing hashes from the output file to prevent duplicate processing and smoothly resume operations. If an uncompressed output file becomes corrupted due to a crash, the tool automatically detects the issue, calculates a fail-safe truncation point, and truncates the file to the last valid sequence before continuing.

Finally, a dry-run mode (`--dry-run` or `-s`) is available to calculate the duplication rate without writing any output files. Since file I/O operations account for the majority of execution time, this mode is exceptionally fast and serves as a highly efficient feature for pipeline planning and data analyses.

3.1 : Algorithmic Complexity and Hardware-Level Optimizations

The architectural principal of `FastDedup`'s performance lies in shifting the deduplication problem from a string-matching paradigm to an integer-matching paradigm. By transforming raw paired-end sequences (e.g., 2×150 bytes) into a single 64-bit hash, reducing the data by 37.5 times in both space and time complexity.

3.1.1 Space Complexity and Memory Footprint

In a deterministic approach, storing a read pair requires retaining sequences of length L (where $L = r_1 + r_2$). The space complexity for N unique sequences scales linearly as $O(N \times L)$. For a standard 150 bp paired-end dataset, each pair consumes at least 300 bytes of raw character data, plus substantial data structure overhead (e.g., string pointers). Storing 10^8 sequences typically exceeds 32 GB of RAM. By computing a 64-bit hash using `xxh3`, `FastDedup` reduces the sequence identity into exactly 8 bytes. This decouples the memory requirement from the read length, reducing the space complexity to $O(N)$ (since $8 \ll L$).

3.1.2 Time Complexity and Arithmetic Logic Unit (ALU) Efficiency

String-based deduplication necessitates sequential, byte-by-byte comparisons. In the worst-case scenario (e.g., highly similar sequences or long identical prefixes), comparing two sequences requires $O(L)$ time. Conversely, while calculating the initial hash takes $O(L)$ time during the single-pass file reading, all subsequent identity verifications are reduced to integer operations. Comparing two 64-bit integers is executed natively by the CPU's ALU in exactly 1 clock cycle, effectively reducing the hash lookup and comparison time complexity to $O(1)$.

3.1.3 Cache Locality and Zero-Allocation Pipeline

Beyond theoretical Big-O improvements, the 8-byte hash architecture heavily exploits modern CPU hardware design. Processors fetch data from RAM in discrete 64-byte cache lines. A 300-byte sequence spans multiple cache lines, leading to frequent cache misses and severe memory bandwidth bottlenecks. In contrast, exactly eight 64-bit hashes fit perfectly into a single cache line. As `fxhash` probes the internal hash set for collisions, it operates almost exclusively within the ultra-fast CPU cache (L1/L2), minimizing latency and maximizing throughput.

Furthermore, `FastDedup` processes byte slices directly from the `needletail` [18] parser into the `xxh3` hasher without ever allocating the raw string to the heap. This zero-allocation pipeline completely bypasses system-level memory allocation (`malloc`) overheads, ensuring that the execution time is strictly bound by disk I/O rather than RAM latency.

3.2 : The Mathematics of Dynamic Hashing

While `FDedup` defaults to the highly efficient 64-bit hashing strategy of `xxh3`, it dynamically assesses the risk of hash collisions to guarantee data integrity. The theoretical risk of a hash collision p for x sequences follows the standard Birthday Problem approximation [19,20]:

$$p = \frac{x^2}{2^{N+1}}$$

Where $N \in \{64, 128\}$ represents the bit-length of the hash. At the default 64-bit configuration and a probability threshold of 0.001, it requires approximately $0.19 * 10^9$ sequences to reach a $1‰$ collision risk. If the estimated dataset size pushes the collision probability beyond this threshold, `FDedup` automatically scales to 128-bit hashing, effectively nullifying the risk (requiring $0.82 * 10^{18}$ sequences to reach the same probability).

For paired-end reads, maintaining this theoretical probability requires bitwise operations. Let S_1 and S_2 represent the nucleotide sequences of Read 1 and Read 2, and H be our non-cryptographic uniform hash function mapping to $\{0,1\}^N$. We define the independent hashes for each read as:

$$h_1 = H(S_1)$$

$$h_2 = H(S_2)$$

If the algorithm combined these hashes using a simple bitwise XOR ($\oplus$), a critical failure occurs when Read 1 and Read 2 are identical sequences ($S_1 = S_2$). In this case, $h_1 = h_2$, resulting in:

$$h_c = h_1 \oplus h_1 = 0$$

Every paired-end read with identical forward and reverse sequences would map to 0. This would cause massive, deterministic hash collisions, meaning the collision probability would no longer be bounded by the theoretical uniform distribution, but rather by the biological frequency of identical sequence pairs in the dataset.

To prevent symmetrical cancellation, the algorithm introduces a bitwise left-rotation by $\frac{N}{2}$ bits, denoted as $R(x, k)$. The unified fingerprint h_u is instead calculated as:

$$h_u = h_1 \oplus R\left(h_2, \frac{N}{2}\right)$$

Under this operation, even if $S_1 = S_2$:

$$h_u = h_1 \oplus R\left(h_1, \frac{N}{2}\right) \neq 0$$

Because $R\left(h_1, \frac{N}{2}\right)$ shifts the lower half of the bits to the upper half and vice versa, $h_1 \oplus R\left(h_1, \frac{N}{2}\right)$ will almost never equal 0. It would only equal 0 if h_1 consisted of two perfectly identical halves, an event with an astronomically low probability of $2^{-\frac{N}{2}}$.

Since a bitwise rotation is a bijective function over the N-bit vector space, it perfectly preserves the original entropy of h_2. When two independent, uniformly distributed random variables are combined using a bitwise XOR, the resulting variable remains uniformly distributed. Therefore, the unified fingerprint h_u safely maintains a strict uniform distribution, ensuring the overall collision risk remains uncorrupted and strictly bounded by the initial probability formula.

This is the mathematical foundation that allows `FDedup` to guarantee a collision probability of less than $1‰$ for datasets containing up to approximately 190 million paired-end reads (380 million total sequences) when using 64-bit hashes.

3.3 : FxHash: Algorithmic Mechanics and ALU Optimizations

While `FastDedup` utilizes `xxh3` to compute sequence fingerprints, it relies on `fxhash` [17] to manage these fingerprints within its internal hash set. Concretely, `FastDedup` uses a `FxHashSet`, a hash set data structure that uses `fxhash` as its internal bucket-routing function: the fixed-size fingerprint produced by `xxh3` is re-hashed by `fxhash` to determine its slot in the table. This two-level design is intentional, `xxh3` provides high-quality, collision-resistant fingerprints of biological sequences, while `fxhash` provides minimal-overhead addressing into the hash table itself.

`fxhash` is a high-performance, non-cryptographic hashing algorithm derived from the internal hasher utilized in the Rust compiler (`rustc`) and Firefox. Its architectural strength lies not in collision resistance, but rather in raw computational throughput and extremely low latency.

For every word extracted from the input, `fxhash` applies a highly optimized state transition function. Let H represent the internal state of the hash (initialized to 0), and W represent an incoming data word:

$$H_{\{i+1\}} = (R_L(H_i, 5) \oplus W_i) \times Kp \bmod \{2^N\}$$

Where:
- N is the machine word size (typically 64 bits).
- $R_L(x, k)$ denotes a bitwise left-rotation of the state x by k bits (`ROTATE = 5`).
- K is a predefined N-bit magic constant (e.g. $K_{64} = 0\text{x}517\text{cc}1\text{b}727220\text{a}95$).

The bitwise left-rotation ensures that previously hashed bits propagate rapidly across the state vector. Being a bijective function, it preserves the entropy of H_i while preventing localized clustering. The XOR then introduces the new word W_i in a single clock cycle, and the wrapping multiplication by the large prime-like constant K acts as an avalanche multiplier, diffusing the newly mixed entropy across all 64 bits. The algorithm explicitly relies on modulo 2^N arithmetic, leveraging the CPU's native handling of integer overflow without any computational penalty.

Within `FastDedup` specifically, `fxhash` receives only the fixed-size fingerprints already produced by `xxh3`, either 64-bit or 128-bit values, corresponding to $L \in \{8, 16\}$ bytes. Because L is bounded to at most 2 machine words, the state transition function executes in exactly 1 or 2 iterations regardless of the input dataset size, making the bucket-routing step a strict $O(1)$ operation. This stands in contrast to the general case, where a byte-oriented hasher processing an arbitrary sequence of length L would require $O(L)$ state transitions.

The algorithm operates entirely in-place on a single internal state variable and requires no auxiliary data structures, lookup tables, or intermediate buffers, yielding an auxiliary space complexity of $O(1)$. Combined with the zero-allocation and cache-locality properties described in Section 3.1, this ensures the memory footprint remains flat during hash table operations, preventing fragmentation and avoiding OS-level memory management overhead.

3.4 : Architecture

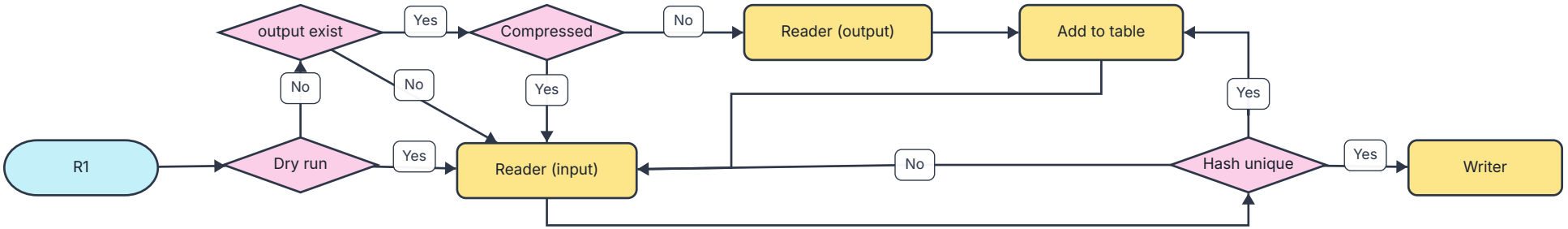


Figure 2: Architectural design of `FastDedup` for single-end reads.

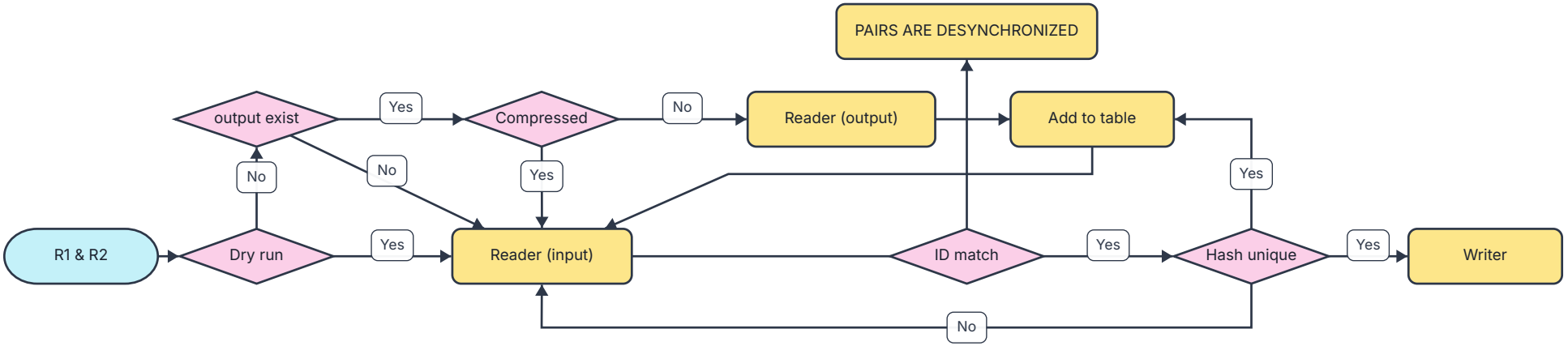


Figure 3: Architectural design of `FastDedup` for Paired-End reads.

4 – Benchmarking and performance evaluation

To test the performances of the differents tools evoked in the previous sections (Table 1), we benchmarked them on a series of synthetic datasets of varying sizes, ranging from 100,000 to 300 million paired-end reads to wich we added 10% of duplicates to each dataset. Thoses reads were generated based of a real human whole-genome (GRCh38) This allows us to evaluate the scalability of each tool and its suitability for different types of sequencing projects, from small-scale experiments to large population genomics studies. To run those benchamrks in a controled environment, we set-up a snakemake pipeline with each tool having 1 core and 33GB of RAM available. This pipeline ran on the High-Performance Computing (HPC) cluster I/O of Meso with AMD EPYC 9654 processors. If the tool goes above 32GB of RAM, it is killed and written as OOM (Out Of Memory) in the results table. In times where RAM and electricity are becoming increasingly costly and environmentally impactful, it is crucial to consider the resource efficiency of bioinformatics tools. Since those tools are for the first step for pre-treatment of data, it would be senseless to run them on ludicrous ressources.

4.1 : Compressed Paired-End FASTQ

The default compression level of `fastp` is 4 but the all the other tools use a compression level of 6, which is the default for `gzip`. Starting at 100M reads, `clumpify` sent an OOM error, disqualifying it for bigger volumes.

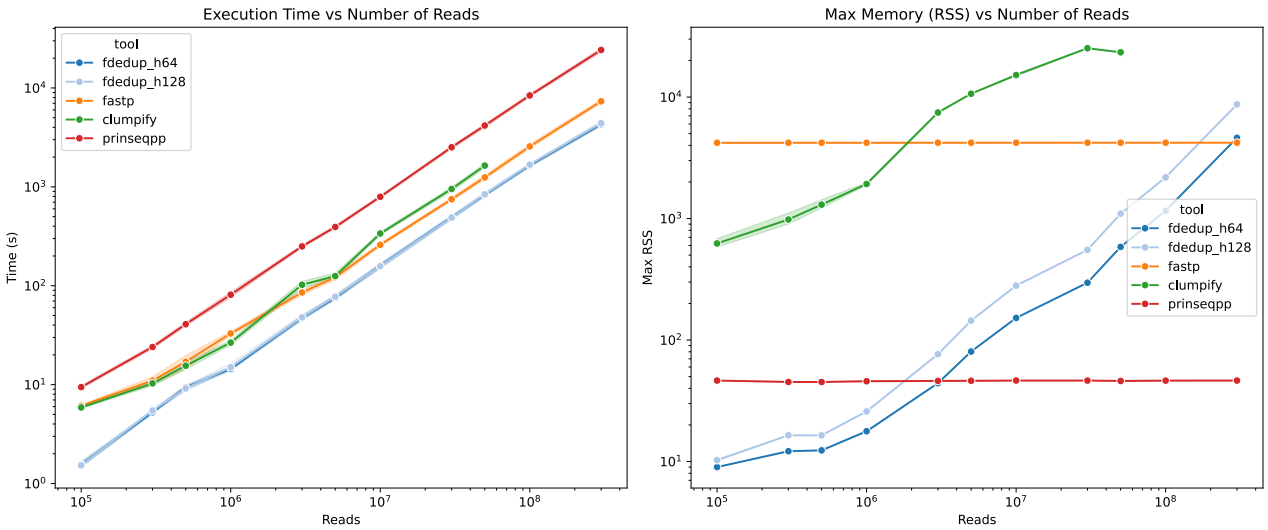


Figure 4: Execution time and memory usage across varying dataset sizes.

After setting `fastp` to use the same compression level, we can see that `FastDedup` is around 50% faster than fastp while using much less RAM until the 300M reads, where it's the roughly same use as fastp for the 64 bit version (Figure 4, Table 4). We can also observed that there are no real difference in execution time between the 64-bit and 128-bit hashing strategies and that the difference of RAM usage bewteen the 64-bit and 128-bit hashing strategies is doubled, which is consistent with the the fact that 64 times 2 is 128.

4.2 : Uncompressed Paired-End FASTQ

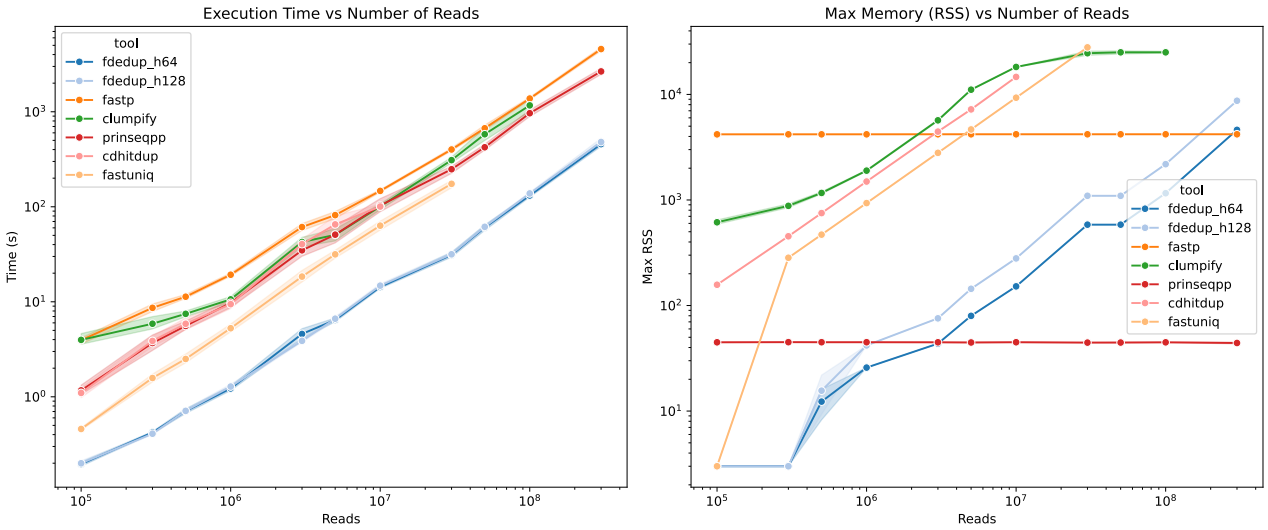


Figure 5: Execution time and memory usage across varying dataset sizes.

For uncompressed FASTQ file as inputs and outputs, `FastDedup` consistently outperformed all other tools in terms of execution time (Figure 5, Table 5). We can also see that `FastDedup` takes up significantly less memory than all the other tools up until 5M reads, where `prinseq++` starts to use less memory than `FastDedup`. However, starting at 30M reads, `CD-HIT-DUP` crashed due to an OOM error, followed by `fastuniq` at 50M reads, and `clumpify` at 300M reads. This left only `fastp`, `prinseq++` and `FastDedup` which was 6.4 times faster than `prinseq++` and more than 10 times faster than `fastp` at 300M reads, while using only 4.6GB and 8.7GB of RAM for 64-bit and 128-bit hashing respectively.

4.3 : Compressed in, decompressed out Paired-End FASTQ

Before talking about the results, we have to note that `fastuniq` and `CD-HIT-DUP` are not included in this benchmark since they do not support compressed files.

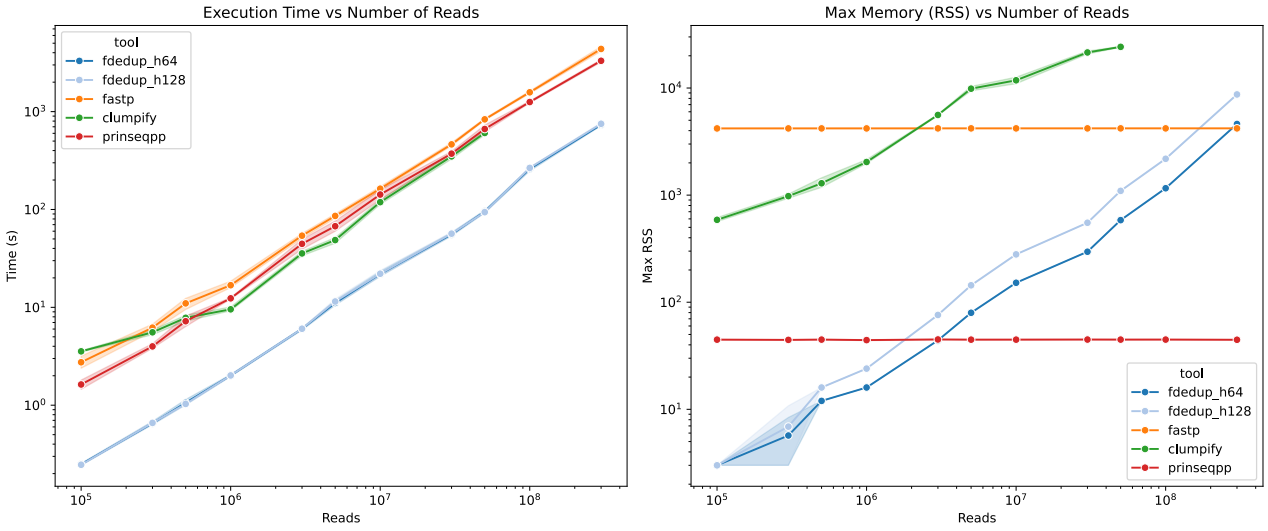


Figure 6: Execution time and memory usage across varying dataset sizes.

We observe very similar results as for uncompressed files (Figure 6, Table 6). `FastDedup` being the fastest tool and using the least amount of RAM up until 5M reads, where `prinseq++` starts to use less memory than `FastDedup`. We can also note very similar results for relative performance between each tools, which mean that all the tools scale in a similar way when the input files are compressed or not.

4.4 : Compressed Single-End FASTQ

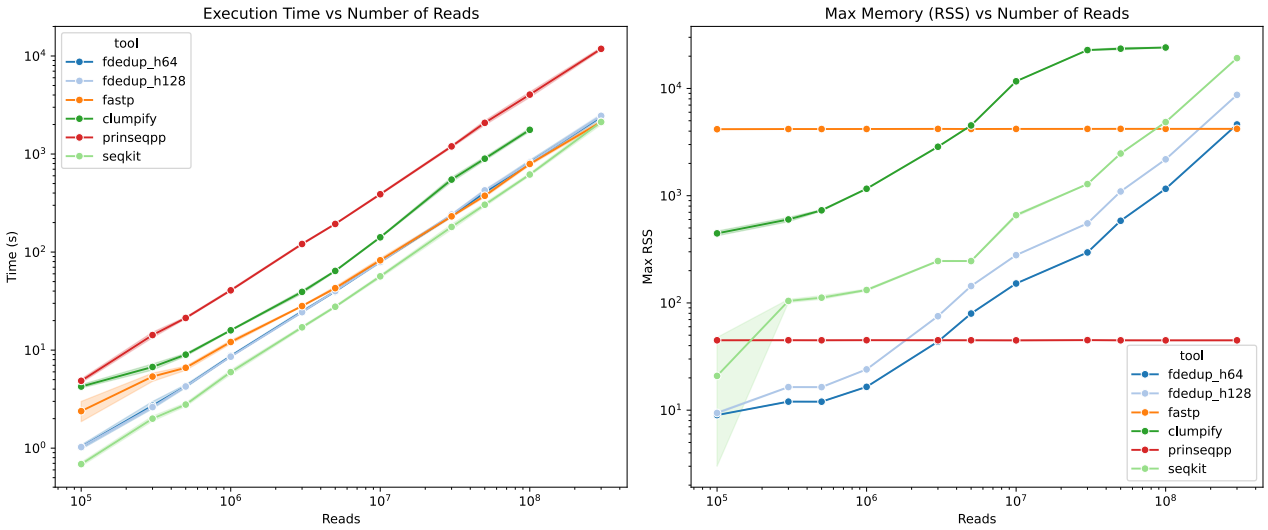


Figure 7: Execution time and memory usage across varying dataset sizes.

For single-end data, we see `SeqKit` as a winner for execution time (Figure 7, Table 7). Despite that `FastDedup` is having much better RAM efficiency than `SeqKit` while being around 25% slower.

4.5 : Decompressed Single-End FASTQ

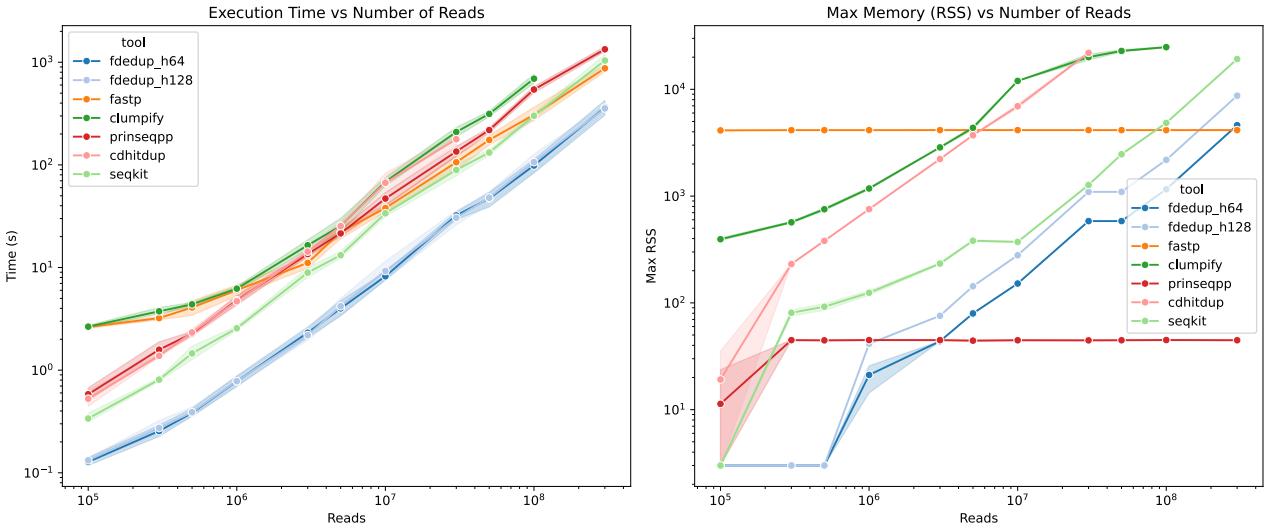


Figure 8: Execution time and memory usage across varying dataset sizes.

`FastDedup` is more than 3 times faster than `SeqKit` and `fastp` at 300M reads while using a quarter of the RAM of `SeqKit` and the same amount as `fastp` (Figure 8, Table 8).

4.6 : Compressed in, decompressed out Single-End FASTQ

We got very similar results than for uncompressed single-end data, with `FastDedup` being the fastest tool for all volumes and using the least amount of RAM up until 5M reads, where `prinseq++` starts to use less memory than `FastDedup` (Figure 9, Table 9).

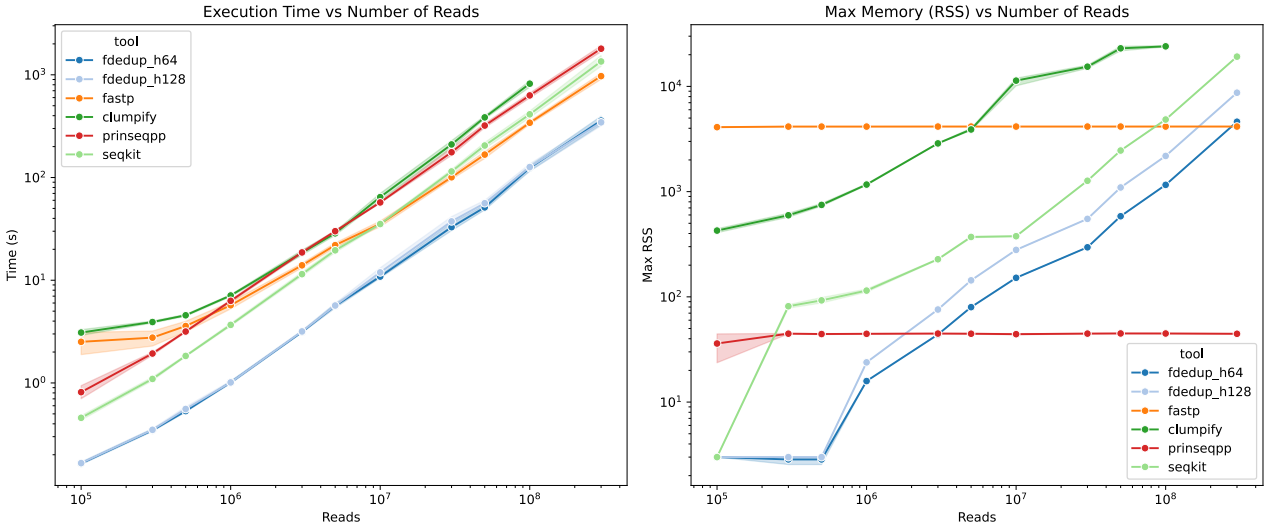


Figure 9: Execution time and memory usage across varying dataset sizes.

5 – Discussion and future perspectives

Now that we have significant data from isolated benchmarks, we can discuss the implications of these results for real-world applications and the future development of `FastDedup`.

5.1 : Output Quality and Deduplication Accuracy

Despite each tool having very different approaches to deduplication, they all produced the pretty much the same output in terms of the number of unique sequences and the duplication rate. There is a small difference, but less than 0.1% between the tools. However, we were not able to estimate for `CD-HIT-DUP` since it doesn't output any log. Moreover, there is a problem with `prinseq++` starting at 10M reads where the duplication rate is higher than 10.1%, reaching up to more than 90% at 300M reads (Figure 10). Because of this we do not recommend using `prinseq++` in your pipelines, since it starting to be more memory efficient than `FastDedup` at 5M reads, but starts having a very high false-positive rate at 10M reads.

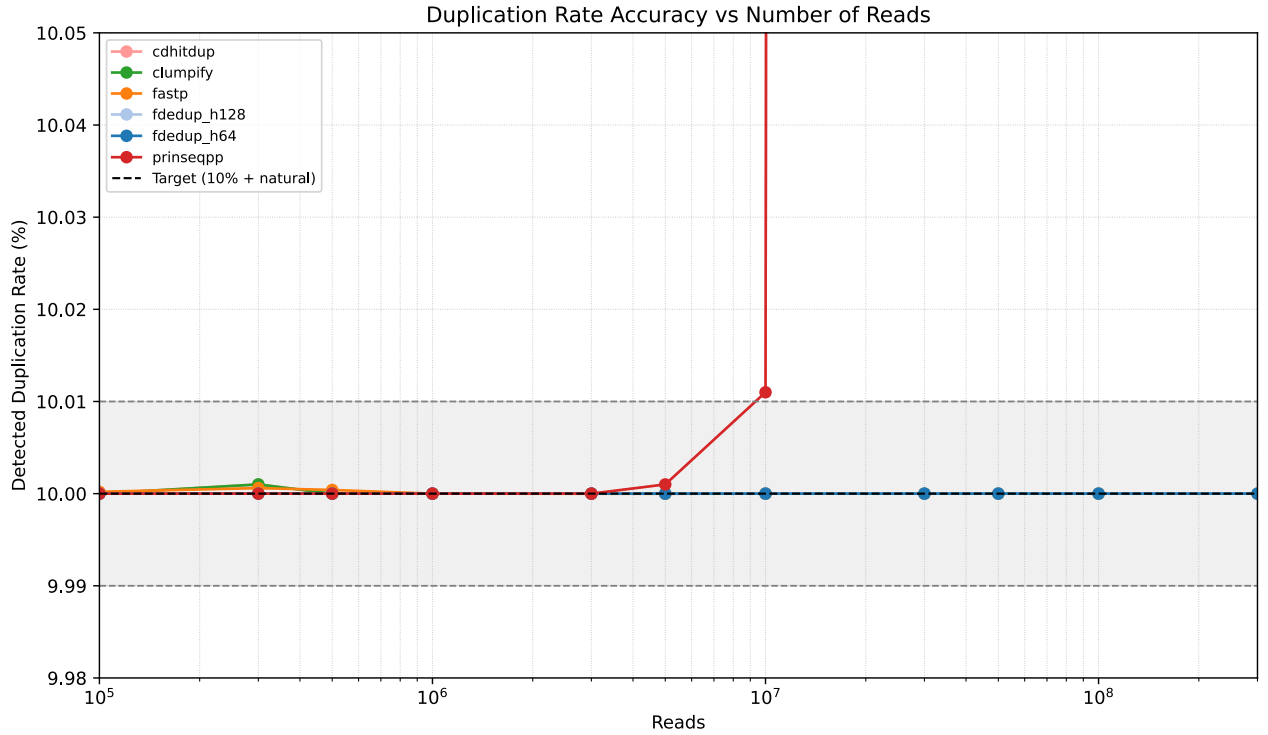


Figure 10: Execution time and memory usage across varying dataset sizes.

A similar pattern is observed for `clumpify` on signle-end data, but starting at 3M reads. It is above 10.1% at 10M reads but reaching around than 10.1% at 100M reads (Figure 11). This is still above the 0.1% false-positive threshold, but it is far better than the 90% false-positive rate of `prinseq++` at 300M reads.

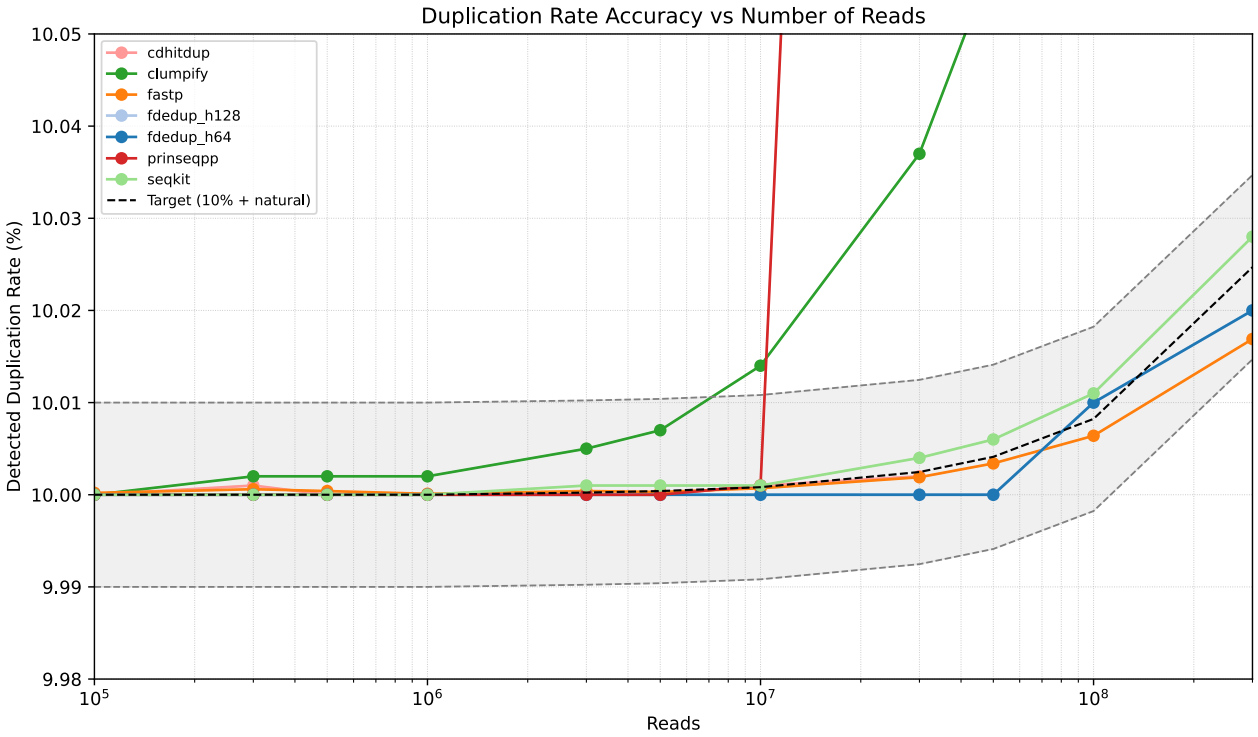


Figure 11: Execution time and memory usage across varying dataset sizes.

The target duplication value isn't constant across different volume because `wgsim` is a random read simulator. We can calculate the number of natural duplicate C with N randomly generate reads from a uniform distribution of M possible start positions like $C \approx \frac{N^2}{2M}$. However, the human genome is not uniform, it contains high-frequency repeat families [21], so we have to aproximate M with the number of unique start positions in the simulated dataset. We can aproximate $k = (2G)^{-1}$ using a known data point from your `wgsim` simulation. Looking at the generation of $270,000,000$ single-end reads, the simulation naturally produced roughly $60,000$ coincidental duplicates.

We can solve k like:

$$60,000 = k \times (270,000,000)^2$$

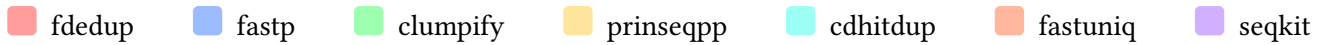
$$k = \frac{60,000}{(270,000,000)^2}$$

We can simply now express the number of natural duplicates C like $C = k \times N^2$ and add this C to the 10% of duplicates we added to the dataset to get the expected duplication rate for each dataset size.

5.2 : In short

Tool	Algorithm / Approach	PE Support	Error Tolerance	Memory Usage	Key Limitations
`fastp` [3,4]	Probabilistic (Bloom filter/hashes)	Yes	No	Configurable	False-positive rate; long runtime
`FastUniq` [5]	Deterministic (Merge sort)	Yes	No	Very High	Severe RAM bottleneck; sensitive to input file order.
`Clumpify` [7]	k-mer grouping	Yes	Yes	Variable	Variable stability on massive datasets; high memory overhead.
`SeqKit rmdup` [8,9]	Exact matching (Hashing/MD5)	No	No	Very Low	Desynchronizes paired-end files.
`CD-HIT-DUP` [10,11]	Greedy incremental clustering	Yes	Yes	Very High	Stability issues in containerized environments.
`PRINSEQ++` [12]	Probabilistic (Bloom filter)	Yes	No	Low	False-positives rate
`FastDedup`	Exact matching (Hashing/xxh3)	Yes	No	Very Low	Theoretical negligible collision risk

Table 1: Summary of existing tools for reads deduplication and their limitations.



Mode	0.1 M			0.3 M			0.5 M			1 M			3 M			5 M			10 M			30 M			50 M			100 M			300 M		
	cc	dd	cd	cc	dd	cd	cc	dd	cd	cc	dd	cd	cc	dd	cd	cc	dd	cd	cc	dd	cd	cc	dd	cd	cc	dd	cd	cc	dd	cd	cc	dd	cd
PE																																	
SE																																	

Table 2: Fastest speed of each tool across different dataset sizes and file formats.

Mode	0.1 M			0.3 M			0.5 M			1 M			3 M			5 M			10 M			30 M			50 M			100 M			300 M		
	cc	dd	cd	cc	dd	cd	cc	dd	cd	cc	dd	cd	cc	dd	cd	cc	dd	cd	cc	dd	cd	cc	dd	cd	cc	dd	cd	cc	dd	cd	cc	dd	cd
PE																																	
SE																																	

Table 3: Lowest RAM of each tool across different dataset sizes and file formats. Prinseq++ is excluded from this table because of the duplication rate issue.

5.3 : The Future of FastDedup

The current version of `FastDedup` is optimized for speed and memory efficiency, but there are some features that could further enhance its functionality. Firstly, the support for multithreading, which would allow the tool to take advantage of multiple CPU cores and further reduce execution time, especially for larger datasets. We can use thoses mutiple cores for parallelizing the file reading and hashing process, while maintaining a shared hash set for deduplication. This can however introduce some overhead due to thread synchronization, so it would require careful testing to ensure that the performance benefits outweigh the costs. However, those extra cores could be used for faster file compression and decompression. Indeed, writing a compressed file makes the writing phase taking more than 90% of the execution time.

Another feature that could be added is track of abundance, which would allow users to not only remove duplicates but also to quantify how many times each unique sequence was observed in the input dataset. This can be particularly useful for applications like metagenomics or transcriptomics, where the abundance of certain sequences can provide important biological insights.

6 – Conclusion and Recommendations

`FastDedup` represents a significant imporvment for FASTX deduplication, combining unmatched speed with a remarkably low memory footprint across all tested scenarios. Its design, shifting the deduplication problem from string-matching to integer-matching through non-cryptographic hashing, allows it to scale from small experimental datasets to the largest population-scale whole-genome sequencing projects without changing configuration or requiring HPC-class hardware.

6.1 : Summary of findings

For paired-end data, `FastDedup` is the clear winner on both dimensions. It consistently outperforms all competitors in execution time, being up to 6× faster than `prinseq++` and more than 11× faster than `fastp` on uncompressed 300M-read datasets, while keeping its memory footprint well below that of every other tool that survived to large volumes. `FastUniq` and `CD-HIT-DUP` are eliminated by memory constraints before reaching 100M reads, and `Clumpify` crashes at 100M reads under the 32GB RAM budget regardless of format. `PRINSEQ++`, despite being the most memory-efficient competitor, disqualifies itself on correctness grounds: its Bloom filter false-positive rate explodes beyond 10% starting at 10M reads, reaching above 90% at 300M reads, making it unsuitable for any analysis where duplicate removal must be reliable.

For single-end data, the picture is more nuanced. `SeqKit` outperforms `FastDedup` on speed when both input and output are compressed, owing to its mature multi-threaded compression pipeline. However, `FastDedup` reverses this advantage completely on uncompressed I/O, where it runs more than 3.5× faster than both `SeqKit` and `fastp` at 300M reads, and maintains a drastically lower memory footprint throughout: at 300M reads, `SeqKit` consumes approximately 19 GB of RAM compared to `FastDedup`'s 4.6 GB, a fourfold difference. Users working on memory-constrained nodes or seeking maximum throughput through uncompressed intermediate files should therefore prefer `FastDedup` even for single-end libraries.

6.2 : Practical recommendations

For maximum performance in any pipeline, we recommend the following strategy: run `FastDedup` immediately after quality control, and direct its output to an uncompressed temporary file. Writing uncompressed output eliminates the compression bottleneck that otherwise dominates execution time, and the temporary files can be recompressed or deleted once downstream steps are complete. For users whose storage constraints require compressed outputs throughout, the compressed-output mode of `FastDedup` remains faster than all alternatives for paired-end data, and is still competitive for single-end data except at the largest scales where `SeqKit` leads on speed alone.

The 64-bit hashing mode is appropriate for the overwhelming majority of datasets: the collision probability remains below 1‰ for libraries of up to approximately 190 million paired-end reads (380 million total sequences). For larger datasets, the automatic upgrade to 128-bit hashing provides the same pipeline interface

with negligible runtime overhead and approximately double the memory cost, a trade-off that remains highly favourable compared to any deterministic alternative.

We advise against `PRINSEQ++` and `clumpify` in any production pipeline handling more than 5M reads. Their false-positive rate are not a theoretical concern but an empirically observed failure: it removes correct unique sequences at increasing rates as dataset size grows, silently corrupting downstream variant calling and assembly results. `CD-HIT-DUP` and `FastUniq` remain valid choices only for small, uncompressed datasets where exact error tolerance or mismatch tolerance are required and where memory is not a constraint.

6.3 : Limitations and future work

The current version of `FastDedup` is strictly single-threaded. While this simplifies the zero-allocation pipeline and guarantees deterministic output ordering, it leaves performance on the table for users with access to multiple cores. The most impactful near-term improvement would be parallelising the compression and decompression stages, which account for more than 90% of wall-clock time in compressed-output modes. Parallelising the hashing and hash-set lookup stages would offer additional gains, though this requires careful handling of concurrent access to the shared hash set.

A second avenue for development is abundance tracking: rather than discarding duplicate records entirely, `FastDedup` could optionally record the copy number of each unique sequence in the output headers. This would extend its utility to metagenomics and transcriptomics workflows where abundance information carries biological meaning.

Finally, the current benchmarking suite covers synthetic data derived from a single human reference genome. Extending the evaluation to metagenomic, amplicon, and low-complexity datasets would provide a more complete picture of performance across the full range of sequencing applications.

`FastDedup` is available under the MIT License at:
- GitHub https://github.com/RaphaelRibes/FastDedup
- Bioconda https://anaconda.org/bioconda/fdedup
- Cargo https://crates.io/crates/fastdedup

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8 – Annexes

Tool *(s)* *(MB)*	**0.1M**	**0.3M**	**0.5M**	**1.0M**	**3.0M**	**5.0M**	**10.0M**	**30.0M**	**50.0M**	**100.0M**	**300.0M**
`clumpify`	5.9±0.2 624±86	10.3±0.9 982±192	15.5±2.0 1302±178	26.6±2.4 1924±71	102±13.8 7458±130	125±13.4 10650±15	337±20.0 15185±469	952±74.4 25226±40	1639±125 23350±257	OOM	OOM
`fastp`	6.1±0.2 4202±0	11.0±1.6 4205±1	17.0±4.3 4207±1	32.9±2.5 4203±1	85.5±7.7 4208±1	122±13.6 4208±1	259±15.3 4209±1	750±49.2 4210±1	1246±102 4209±1	2564±195 4209±1	7324±537 4210±1
`fdedup_h128`	1.5±0.1 10±0	5.5±0.3 16±0	9.2±0.9 16±0	15.0±2.1 26±0	47.9±4.2 76±0	77.5±4.8 144±0	158±12.1 281±0	489±31.9 552±0	841±74.9 1096±0	1679±90.5 2184±0	4407±343 8712±1
`fdedup_h64`	1.6±0.2 9±0	5.3±0.3 12±0	9.4±0.8 12±0	14.4±1.1 18±0	47.2±4.3 44±0	75.4±6.6 80±0	159±15.1 152±0	493±34.8 296±1	830±44.0 584±1	1649±90.9 1160±0	4292±323 4616±1
`prinseqpp`	9.4±0.6 46±0	24.0±1.5 45±0	40.8±3.0 45±0	81.2±7.8 46±1	250±13.2 46±0	393±23.9 46±0	793±39.3 46±0	2515±146 46±0	4169±257 46±1	8399±441 46±0	24202±1657 46±0

Table 4: Execution time and memory usage across varying dataset sizes for compressed input and compressed output paired-end FASTQ files.

Tool *(s)* *(MB)*	**0.1M**	**0.3M**	**0.5M**	**1.0M**	**3.0M**	**5.0M**	**10.0M**	**30.0M**	**50.0M**	**100.0M**	**300.0M**
`cdhitdup`	1.1±0.2 157±1	3.9±0.8 453±0	5.9±1.3 751±0	9.4±1.6 1495±1	40.4±8.4 4443±1	65.2±22.4 7214±119	101±10.4 14657±37	OOM	OOM	OOM	OOM
`clumpify`	4.0±0.9 615±47	5.9±1.5 880±60	7.5±0.8 1166±45	10.6±1.2 1896±34	42.6±7.8 5683±143	50.6±12.8 11092±150	101±11.6 18211±35	310±33.5 24576±1880	580±147 25061±1308	1167±287 25057±751	OOM
`fastp`	3.9±0.5 4187±0	8.6±1.4 4187±1	11.3±0.9 4188±1	19.3±1.2 4188±1	61.2±8.5 4191±0	81.7±11.1 4192±0	147±7.5 4192±1	401±20.6 4192±0	670±78.2 4192±0	1378±71.8 4192±2	4569±319 4192±1
`fastuniq`	0.5±0.0 3±0	1.6±0.2 284±0	2.5±0.4 470±0	5.3±0.8 935±0	18.4±4.6 2796±0	31.6±4.8 4658±0	63.3±10.9 9312±0	174±26.2 27927±0	OOM	OOM	OOM
`fdedup_h128`	0.2±0.0 3±0	0.4±0.0 3±0	0.7±0.1 16±10	1.3±0.1 42±0	3.9±0.2 76±0	6.6±0.5 144±0	14.8±1.0 280±1	31.6±2.7 1096±0	61.5±4.6 1095±0	138±6.0 2184±0	481±72.7 8712±0
`fdedup_h64`	0.2±0.0 3±0	0.4±0.0 3±0	0.7±0.1 12±6	1.2±0.1 26±0	4.6±0.9 44±0	6.4±0.5 80±0	14.3±0.8 152±0	31.1±3.0 584±1	60.4±6.5 584±0	131±8.7 1159±0	456±47.4 4616±0
`prinseqpp`	1.2±0.2 45±0	3.7±1.2 45±0	5.6±1.0 45±0	9.7±1.5 45±0	34.8±8.7 45±0	50.9±15.8 45±0	103±27.7 45±0	248±34.8 45±0	423±53.3 45±0	965±122 45±0	2658±309 44±1

Table 5: Execution time and memory usage across varying dataset sizes for uncompressed input and uncompressed output paired-end FASTQ files.

Tool *(s)* *(MB)*	**0.1M**	**0.3M**	**0.5M**	**1.0M**	**3.0M**	**5.0M**	**10.0M**	**30.0M**	**50.0M**	**100.0M**	**300.0M**
`clumpify`	3.6±0.1 588±43	5.6±0.5 979±73	7.8±0.9 1290±212	9.5±0.8 2039±160	35.6±3.0 5604±94	48.7±5.1 9863±755	119±10.0 11807±1321	349±39.7 21482±1110	605±48.2 24234±345	OOM	OOM
`fastp`	2.7±0.6 4196±1	6.2±0.9 4197±1	11.0±2.3 4197±1	16.8±2.4 4197±1	54.0±4.7 4200±0	85.9±6.9 4200±0	163±12.9 4200±1	463±28.8 4200±0	834±30.6 4200±1	1576±71.0 4200±1	4369±399 4201±0
`fdedup_h128`	0.2±0.0 3±0	0.7±0.0 7±6	1.0±0.0 16±0	2.0±0.1 24±0	6.0±0.1 76±0	11.5±1.1 144±0	22.1±2.6 280±1	56.6±3.1 552±0	93.9±3.6 1096±0	266±4.8 2184±0	751±32.9 8712±0
`fdedup_h64`	0.2±0.0 3±0	0.7±0.0 6±4	1.1±0.1 12±0	2.0±0.1 16±0	6.0±0.2 44±0	11.1±1.0 80±0	21.9±2.0 152±0	55.7±3.6 296±0	95.3±4.9 584±0	260±17.0 1160±0	734±28.1 4615±0
`prinseqpp`	1.6±0.3 45±0	4.0±0.4 45±0	7.2±1.5 45±0	12.4±0.3 44±1	44.6±8.4 45±0	67.5±13.3 45±0	142±27.2 45±0	372±31.9 45±0	664±95.4 45±0	1253±44.5 45±0	3306±161 45±0

Table 6: Execution time and memory usage across varying dataset sizes for compressed input and uncompressed output paired-end FASTQ files.

Tool *(s)* *(MB)*	**0.1M**	**0.3M**	**0.5M**	**1.0M**	**3.0M**	**5.0M**	**10.0M**	**30.0M**	**50.0M**	**100.0M**	**300.0M**
`clumpify`	4.2±0.3 445±36	6.7±0.8 601±46	9.0±0.5 730±19	15.9±0.4 1161±9	39.3±2.6 2860±38	64.2±1.5 4520±19	141±3.4 11665±20	549±48.0 22803±148	897±62.6 23497±923	1766±112 24092±140	OOM
`fastp`	2.4±0.9 4171±2	5.4±0.8 4185±3	6.6±0.6 4185±1	12.1±0.9 4189±3	28.1±0.5 4197±2	42.9±2.3 4189±2	82.5±6.8 4195±4	231±11.0 4197±2	375±24.3 4201±3	792±45.9 4200±1	2126±145 4202±1
`fdedup_h128`	1.0±0.1 9±0	2.6±0.1 16±0	4.3±0.0 16±0	8.6±0.1 24±0	24.4±0.5 75±0	40.1±0.9 144±0	79.3±2.7 279±0	241±5.9 552±0	426±18.0 1095±1	815±66.3 2184±0	2452±213 8712±0
`fdedup_h64`	1.0±0.1 9±0	2.7±0.4 12±0	4.3±0.3 12±0	8.7±0.3 17±0	24.7±1.1 44±0	39.7±0.8 80±0	80.1±3.6 152±0	234±10.3 296±0	406±26.7 584±0	819±67.6 1160±0	2394±172 4616±0
`prinseqpp`	4.9±0.3 45±0	14.2±1.5 45±0	21.3±0.8 45±0	40.7±1.4 45±0	121±3.5 45±0	194±2.9 45±0	389±12.3 45±0	1197±47.4 45±0	2080±182 45±0	4038±414 45±0	11827±670 45±0
`seqkit`	0.7±0.1 21±36	2.0±0.2 104±6	2.8±0.2 112±7	6.0±0.5 132±5	17.1±0.8 246±4	27.7±1.1 246±3	56.6±3.3 659±30	180±17.9 1284±3	304±21.6 2474±6	618±32.1 4859±6	2123±214 19184±26

Table 7: Execution time and memory usage across varying dataset sizes for compressed input and compressed output single-end FASTQ files.

Tool *(s)* *(MB)*	**0.1M**	**0.3M**	**0.5M**	**1.0M**	**3.0M**	**5.0M**	**10.0M**	**30.0M**	**50.0M**	**100.0M**	**300.0M**
`cdhitdup`	0.5±0.2 19±32	1.4±0.1 231±0	2.3±0.3 381±0	4.7±0.8 754±1	14.3±1.8 2223±1	25.4±3.6 3714±1	67.0±23.6 6943±469	178±37.9 21958±230	OOM	OOM	OOM
`clumpify`	2.7±0.1 395±18	3.7±0.5 568±18	4.4±0.3 753±37	6.2±0.5 1179±45	16.5±3.2 2858±35	25.4±6.7 4360±293	68.5±10.6 11985±30	209±37.8 20013±2129	314±31.3 22873±809	692±116 24829±469	OOM
`fastp`	2.6±0.1 4125±29	3.2±0.2 4147±1	4.1±1.0 4147±1	6.0±1.2 4147±1	11.1±2.3 4150±1	22.1±5.2 4150±1	38.0±4.7 4149±0	106±27.6 4149±1	175±30.4 4149±0	308±73.5 4149±1	873±117 4150±1
`fdedup_h128`	0.1±0.0 3±0	0.3±0.1 3±0	0.4±0.1 3±0	0.8±0.1 41±1	2.2±0.4 76±0	4.2±1.2 143±1	9.3±3.2 279±0	30.5±8.0 1095±0	47.9±10.6 1095±0	106±32.9 2183±1	355±47.1 8711±1
`fdedup_h64`	0.1±0.0 3±0	0.3±0.1 3±0	0.4±0.1 3±0	0.8±0.1 21±9	2.3±0.4 44±0	4.0±1.1 80±0	8.2±1.2 152±0	32.2±4.5 583±0	47.0±12.1 583±0	98.6±23.8 1159±0	363±96.2 4615±0
`prinseqpp`	0.6±0.1 11±17	1.6±0.4 45±0	2.3±0.2 45±1	4.8±0.8 45±0	13.5±1.3 45±0	21.5±2.7 44±1	47.0±10.8 45±0	135±25.7 45±0	218±18.7 45±0	543±64.1 45±0	1337±123 45±0
`seqkit`	0.3±0.1 3±0	0.8±0.0 81±9	1.5±0.4 92±10	2.6±0.2 124±10	8.9±1.7 233±7	13.2±1.6 382±11	33.8±4.5 372±6	89.3±21.0 1271±10	132±13.5 2463±8	301±29.8 4848±9	1040±213 19206±27

Table 8: Execution time and memory usage across varying dataset sizes for uncompressed input and uncompressed output single-end FASTQ files.

Tool *(s)* *(MB)*	**0.1M**	**0.3M**	**0.5M**	**1.0M**	**3.0M**	**5.0M**	**10.0M**	**30.0M**	**50.0M**	**100.0M**	**300.0M**
`clumpify`	3.1±0.3 426±30	3.9±0.1 596±43	4.6±0.2 748±37	7.1±0.3 1167±27	18.5±1.6 2873±28	28.6±2.1 3896±150	64.5±8.8 11344±1820	210±26.6 15394±838	385±21.2 22998±1757	820±93.9 24023±577	OOM
`fastp`	2.5±1.0 4100±70	2.8±0.8 4151±1	3.6±0.7 4151±0	5.7±0.8 4151±1	14.0±1.1 4154±1	22.0±1.8 4154±1	35.4±3.7 4153±1	101±8.9 4153±0	167±21.3 4153±1	341±20.0 4154±2	971±109 4154±2
`fdedup_h128`	0.2±0.0 3±0	0.3±0.0 3±0	0.6±0.0 3±0	1.0±0.0 24±0	3.2±0.1 76±0	5.7±0.4 143±1	11.9±1.9 279±0	37.4±6.6 551±0	56.2±6.3 1095±0	126±17.0 2183±1	344±42.8 8711±1
`fdedup_h64`	0.2±0.0 3±0	0.3±0.0 3±0	0.5±0.0 3±0	1.0±0.0 16±0	3.1±0.1 44±0	5.6±0.4 80±0	10.9±0.6 152±0	32.9±5.1 296±0	51.2±4.4 584±0	124±12.1 1159±0	358±58.2 4616±0
`prinseqpp`	0.8±0.2 36±17	1.9±0.1 45±0	3.2±0.1 44±1	6.3±0.3 45±0	18.7±1.0 45±0	30.0±0.9 45±1	57.2±1.5 44±1	176±24.9 45±0	321±25.4 45±0	630±55.4 45±0	1796±206 45±0
`seqkit`	0.5±0.0 3±0	1.1±0.1 81±5	1.8±0.1 93±10	3.7±0.2 115±6	11.5±0.8 228±3	19.6±1.4 371±9	35.2±1.4 377±5	115±8.0 1268±4	205±22.6 2457±5	413±50.6 4843±7	1349±255 19182±32

Table 9: Execution time and memory usage across varying dataset sizes for compressed input and uncompressed output single-end FASTQ files.